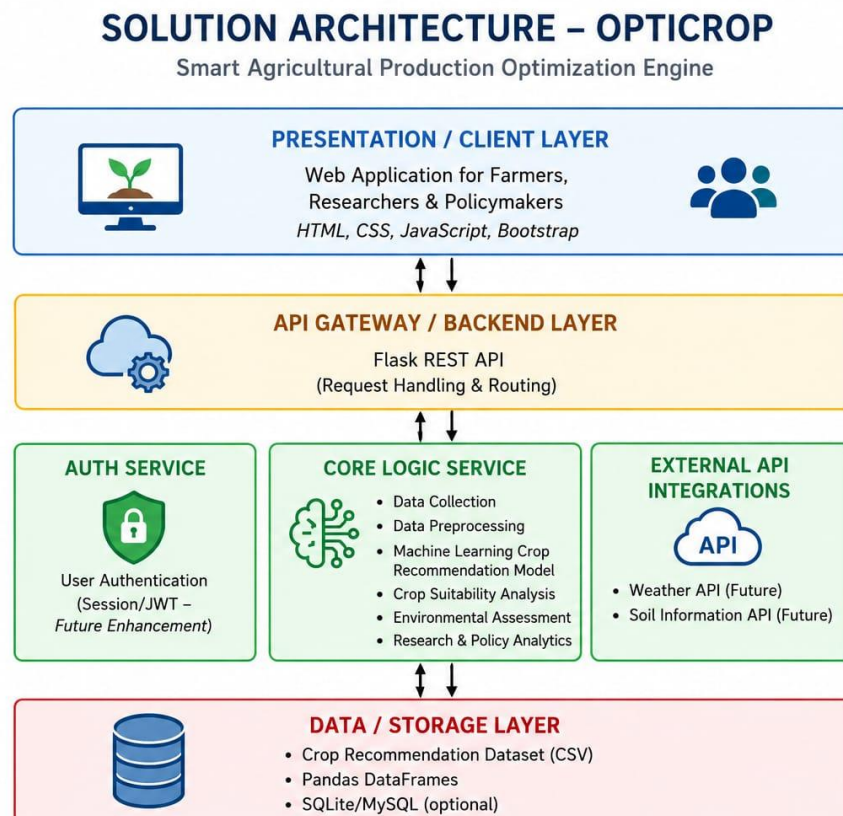


Solution Architecture

Date	01 July 2026
Team ID	XXXXXX
Project Name	OptiCrop: Smart Agricultural Production Optimization Engine
Maximum Marks	5 Marks

Solution Architecture Diagram:

Provide a high-level visual representation of your solution's architecture. Use the structural layout below as a template to map out your Presentation Layer, Application Layer, and Data Layer. Fill in the specific technologies and components for your project directly into the boxes to create your architectural diagram.



Component Description Table

Component Name	Description / Role in Architecture	Technologies Used
Presentation Layer	Collects soil inputs (N, P, K, pH, temperature, humidity, rainfall) and displays crop recommendations.	HTML, CSS, JavaScript, Bootstrap
API Gateway	Routes requests from the client to the backend services.	Flask REST API
Core Logic Service	Performs event analysis, topic extraction, conversation generation, history logging, feedback collection, and fact verification	Python, Scikit-learn, Pandas, NumPy
Database / Storage	Stores crop dataset, prediction data and application records.	CSV Dataset, SQLite/MySQL